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The Health of Sexual Minorities

*Public Health Perspectives on
Lesbian, Gay, Bisexual
and Transgender Populations*

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Sampling in Surveys of Lesbian, Gay, and Bisexual People

Diane Binson, Johnny Blair, David M. Huebner, and William J. Woods

1 Introduction

One purpose of this volume is to provide methodological tools for conducting public health research for lesbian, gay, and bisexual (LGB) populations. Among the most fundamental methodological considerations in any kind of research is how best to sample the population of interest. The importance of sampling in health research among LGB people can be seen in how the medical profession initially came to consider homosexuality a mental disorder. The “evidence” supporting such judgments came from studies of psychiatrists’ patients and inmates in mental hospitals and prisons. The possibility that such samples might be biased—that is, that these individuals might not be representative of homosexuals who were not in treatment or institutionalized—was not considered seriously, in large part due to prevailing attitudes about homosexuality. Nevertheless, the historical lesson of how poor sampling can create significant problems for the health and well-being of LGB people should be enough to make all of us ardent promoters of the use of sound sampling procedures in public health research.

This chapter addresses sampling broadly but focuses primarily on procedures for probability sampling used in survey research. Although nonprobability sampling is discussed and has proved useful in particular circumstances, it should be stated clearly that the lack of representativeness, as one has no basis for knowing if they are representative, makes them significantly less useful when describing the larger LGB population. Public health has been seriously underserved by the lack of statistically defensible information describing the health behaviors and needs of the various LGB populations.

This lack of information on LGB health is not due to an absence of population studies. National government agencies have conducted health-related studies of the U.S. population using sophisticated sample designs for decades. The National Health Interview Survey (NHIS), which monitors the “health of the nation,” has been conducted since 1957. Without asking even basic questions about sexual behavior,

identification, or partnership on such population surveys, it was not possible to estimate accurately the number of LGB persons in the U.S. population or to identify their unique health care needs.

Kinsey and colleagues (1948, 1953) conducted extensive studies of human sexual behavior, but it is well known that there were fundamental problems with the sample of men and women on whom Kinsey and colleagues reported. For example, we know that the participants were made up of volunteers; as volunteers, one would expect that they would be somewhat more comfortable discussing and disclosing aspects of their sexuality and might well be more sexually experienced. Additionally, we know that the sample in the Kinsey group's 1948 volume on men included respondents drawn from sources that would have increased their likelihood of having had homosexual experiences, in particular, men from all-male institutions such as reform schools, jails, and prisons as well as men drawn from homosexual social networks. This has been discussed at some length by one of Kinsey's major collaborators (Gebhard, 1972; Gebhard & Johnson, 1979). Gebhard made a reasoned estimate on the prevalence of male homosexuality based on a cleaned-up sample and using somewhat different criteria. He concluded that "about 4% of the white college-educated males are predominately homosexual" over the course of their lives (Gebhard, 1972, p. 27).

Sadly, it took a health crisis among gay men to acknowledge the need for basic population data on LGB communities. When circumstances of the acquired immunodeficiency syndrome (AIDS) epidemic called for information on the patterns of sexual behavior of the U.S. population—information needed to design, implement, and evaluate programs to curb the spread of the epidemic—the public health archive of suitable data was virtually empty. In the context of the AIDS epidemic, however, funding was finally made available to draw samples of the general population that would allow prevalence estimates of same-gender sexual contact, as was later funding to draw representative samples of gay men. It was only recently that large federal- and state-level health surveys included questions related to sexual behavior and same-gender sexuality that facilitate analysis of health issues of LGB populations. Without continued resources, both financial and scientific, LGB health studies would be limited to small, nonprobability samples that limit the generalizability of findings (Solarz, 1999). Probability samples can provide reliable estimates and descriptions of health behavior and risks among LGB communities, thereby creating opportunities to design and target more accurately health education and disease prevention efforts that are relevant to the behavioral risks and needs of LGB people.

The rest of the chapter is divided into several parts. First, we provide a rationale for choosing to use probability sample designs and then define basic concepts used for probability sampling. The middle sections describe available probability sampling designs. These are followed by examples of uses of probability sampling in the literature on LGB health. The next section deals with nonprobability sampling, and the circumstances when such sampling is appropriate, the kinds of

nonprobability sampling used, and examples of studies using such procedures. Finally, we provide a short discussion of two areas of sampling that are attracting a lot of attention among researchers: respondent-driven sampling and web-based sampling.

2 Drawing a Probability Sample

Careful sampling begins with the ability to enumerate (i.e., identify and count) the population of interest. If a population can be easily enumerated, it can be sampled in a fairly straightforward manner; to the extent that enumeration is difficult, the sampling is similarly difficult. Many circumstances may make a population difficult to enumerate. Samples of LGB populations or other groups defined by behaviors or self-identification are typically difficult to enumerate for two reasons. First, the population is not identifiable from the sampling frame (as, for example, are residential households in the telephone directory); therefore potential sample members have to be screened and asked to provide information to indicate whether they should be counted as population members. Second, the nature of the information may be considered, by some or most population members, to be sensitive and/or socially undesirable or to entail possible societal risks should it become known. LGB populations are what sampling statisticians would describe as “rare and elusive populations” (Sudman et al., 1988; Solarz, 1999). Although sampling these various rare and elusive populations may present several similar obstacles, each population has unique characteristics that need to be taken into consideration when designing a sample.

2.1 Sampling Rare Populations: Unknown Screening Rates and Costs

Conducting a general population survey of LGB individuals in the U.S. population is a daunting task. In the absence of detailed population information to be used when designing an efficient sample, such a survey would rely mainly or entirely on screening a general-purpose household sample. A massive screening effort would be needed to locate eligible respondents among the general population. Hundreds of thousands of households would have to be contacted before locating potentially eligible persons. Even then, how likely would the potentially eligible persons be to identify as a member of the target population? If they did report eligibility, would they be willing to participate in the survey? Because there are total geographic areas in which the prevalence of the targeted population is unknown and because the screening costs would undoubtedly be expensive, one strategy has been to focus research in cities that tend to have large concentrations of LGB populations and then to screen only households in certain neighborhoods in those cities. In the Urban Men’s Health Study (UMHS) (Catania et al., 2001), for example, telephone numbers in selected ZIP codes were screened for eligibility. Nevertheless, while limiting the geographic area reduced the time and effort needed to

screen households, the process still resulted in a large number of dialed phone numbers. For example, in one city in the UMHS study, 53,050 phone numbers in selected areas of the city were dialed to complete 915 interviews (Pollack, 1999, personal communication). Added to screening costs is the issue that screening rates are unknown and can vary across areas where sampling is being done. Generally when a sample is drawn, the number of households that must be screened to result in a desired number of completed interviews can be estimated using census data or other estimates of the population residing in particular areas. Given the lack of reliable prevalence data of LGB households, how large a sample one needs to start with to end up with “X” number of interviews involves significant guesswork. Starting with too small a sample typically poses a larger problem than beginning with too large a sample, as underestimating the initial sample requires selecting a second sample, which means a serious increase in effort. Hence starting with a larger sample, and then working a subsample from the larger sample to learn the actual screening rates, and using as many subsamples as needed to end up with the projected number of completed interviews is a less costly, more prudent procedure.

2.2 Sampling Elusive Populations: Identification and Response Rate

Another obstacle in sampling LGB populations involves the willingness of potential eligible respondents to be interviewed. Although successfully convincing individuals to participate in an interview is a common problem for most surveys, sampling LGB individuals may also present another difficulty. Some individuals are comfortable telling anyone, strangers and friends alike, that they are gay. However, there may be good reasons (e.g., fear of stigmatization or discrimination) why other individuals would not be willing to tell strangers on the phone that they identify as gay and/or engage in the particular behaviors that would make them eligible to participate in the study. Sampling is further complicated by how researchers may approach defining eligibility and who then is eligible to be in the sample. Sometimes eligibility is defined as how one wants to self-identify; other times it is behavioral, as is the case for most human immunodeficiency virus (HIV)-related studies. To complicate eligibility even further, behavioral definitions may include time boundaries. These challenges make it difficult to compute response rates, as there is so little information on which to base population estimates. Hence, computation of response rates in such situations relies on making certain assumptions about sample members whose eligibility is unknown. For example, when there is less information available than might be wished to estimate the prevalence of the targeted population among “noncontacts” (i.e., households that were selected to be in the sample but when contacted were never home during the entire data collection period of a study), information about these noncontacts remains unknown.

3 Basic Sampling Concepts

Many of the sample designs necessary to sample LGB populations efficiently are specialized and complex. However, all of them depend on understanding a few basic sampling principles and designs. This section provides a brief overview of some fundamental concepts of probability sampling. It is necessary to understand the logic of these common sampling approaches before confronting the more complex issues involved in sampling rare or elusive populations or populations that cannot be easily sampled via household selection. With this background, the next section describes variations and combinations of the basic sample designs to adapt them to LGB populations and includes specialized designs such as network sampling and site or time/location sampling.

3.1 Defining the Population of Interest

Defining the population is the starting point in survey sample design. Survey populations should always be precisely defined. The essence of the population definition has to do with the boundaries set on the group of interest. Whatever definition is settled on, it must be operationalized; that is, there must be a clear set of rules that can be applied to a sampling frame or to individuals who are potential sample members to determine their eligibility for inclusion uniformly. This operationalization sometimes requires compromises that create a slight divergence between the survey population and the (target) population for which the researchers wish to make inferences.

It is important to recognize that a population definition is a construct of the researcher, formulated to meet a particular survey purpose. For one survey, a gay 20-year-old Latino is simply a member of the “general population of adults.” For another survey that person may be part of a target population, “Latino males.” Still another survey may “define” him as part of the target population because he is gay. The survey construct may or may not be how that person thinks of himself. The terms “gay” or “homosexual” in screening questions do not necessarily identify the entire population of “men who have sex with men.” It captures only those sample members who use those terms themselves or are willing to be labeled with them for the purposes of a survey.

The process of defining the population begins to suggest some sampling issues that need to be addressed. If the population is defined as visitors to some venue, questions of access, available lists (frames), or on-site enumeration need to be answered. If the population is defined more generally as a subset of the general population, questions of overall prevalence, differential prevalence by location, and screening response rates must be addressed before a sample design and sample size can be finalized.

3.2 Representative Sample

The goal of a probability sample is to select samples from populations of interest for the purpose of describing characteristics of those

populations, testing models about how their members behave, or assessing the impact of programs and interventions, among other possible analyses. These objectives all involve statistical procedures that assume certain characteristics of the sample, such as randomness and independence of observations. Although a formal consideration of the assumptions of different statistical procedures is beyond the scope of this chapter, it is essential to keep in mind that statistical procedures and results are valid only if the assumptions on which they are based are not seriously violated. Careful analysis of a haphazardly selected sample is pointless.

In an everyday sense, any group of population members, however selected, is a sample of the population. Selection could be by way of volunteer respondents (as when people call a 1-900 telephone number to register their opinion on some issue), people selected by the interviewers' discretion on the street or in a retail establishment, even people selected haphazardly from a telephone book or other list. Such samples *may*, on some level, reflect characteristics of the populations from which they were selected, but we cannot be sure this is true; we have no scientific basis for asserting that samples drawn in these ways are at all "representative" of the target population from which they were taken.

What does *representative* mean? As with many survey terms, representative has multiple everyday meanings. It also has been noted that even in the scientific literature the term has been used in quite different ways. In a review and examination of those uses, Kruskal and Mosteller (1979) took a definition from a sampling text by Stephan and McCarthy (1958) as being closest to what we mean by representative in survey research. The central criterion of this definition is that representative samples are those permitting good estimation.

A representative sample is a sample which, for a specified set of variables, resembles the population . . . [in that] certain specified analyses . . . (computation of means, standard deviations etc.) yield results. . . . within acceptable limits set about the corresponding population values, except that . . . [rarely] the results will fall outside the limits [Stephan & McCarthy, 1958, pp. 31–32].

A couple of aspects of this definition are worth pointing out. First, the "representation" is linked to "specific analyses." Specific analyses imply particular variables (population parameters) of interest. This means that one wants the sample statistic, such as the proportion of people with health insurance coverage, to resemble the same statistic for the population. The researcher has a list of such variables that are operationalized (i.e., defined) by the survey questions corresponding to each variable. The objective is that the survey be representative with respect to this list of variables and their associated analyses.

Second, the definition requires that these sample statistics be close to the corresponding population values "within acceptable limits." This implies that each sample statistic is an "estimate" of the corresponding population statistic (i.e., population parameter). An estimate is not expected to be 100% accurate. It is important to be able to describe how

accurate is a sample estimate. That is, the sample estimates *resemble* the population to an extent that can be quantitatively specified.

It is a common misunderstanding to envision a sample as being either representative or not. Frequently, in discussions of sample size, one hears the criterion that the sample should be large enough that it is “representative of the population.” The matter is more complicated than that. The notion of representativeness is realistically viewed only as a continuum of precision in the context of the survey’s needs.

To have a statistical basis for projecting from a sample back to the population from which it was selected, the sample must be one in which population members are chosen by some random mechanism in such a way that every member has a known, nonzero chance or probability to be included. When these conditions hold, there are laws of statistics that provide a basis for saying that the sample represents the population from which it was chosen. Such a sample is called a *probability sample*. Probability samples are representative of their populations in the Stephan-McCarthy sense. Probability samples also permit estimates of their precision, or *sampling error*.

3.3 Locate or Construct Frames

The *sampling frame* is a set of elements from which a subset is selected during each stage of the process of sampling population members. Ideally, the sampling frame is simply a list that includes all the eligible population members and no nonpopulation members. But a sampling frame may be one or more steps away from the actual sample members. For example, in a telephone survey of the adult population, the primary sampling frame is a list of telephone numbers. These telephone numbers are not themselves sample members. Telephone numbers are the first set of elements sampled *during the process* of reaching population members. The first stage of that process is sampling households. Some of the telephone numbers are residences in which one or more adults reside. Typically, the adults in the household are enumerated. That list is a second sampling frame from which the household (target population) members are selected.

To take one more example, suppose one is sampling visitors to a retail establishment. Ignoring the sample design details for now, one way this might be done is to sample times of day when the establishment is open for business and for each of the selected times sample customers who are present. The primary sampling frame is some list of all the possible times of day, perhaps in 1-hour intervals. A subset of these times is selected. Then, within each selected time, the customers are enumerated. The secondary frame is this list of customers. It is from this frame that target population members (customers) are selected.

The idealized sample frame as a list of all population members and only those population members is rarely achieved, but it is nonetheless important to keep in mind. Actual sample selection begins at this conceptual starting point. The need to have a suitable frame at each stage of selection is also important. Each frame should be a complete listing

of that stage's sample elements; when it is not, the whole design is weakened.

Care must be taken when making use of available frames. First, the frame may have problems in terms of coverage of the population, including some eligibles, omitting others, and including some ineligibles as well. For example, a list of club members may include some who are no longer active and omitting recent registrants. Second, the information in the frame, such as addresses and telephone numbers, may be incorrect. Almost no frame is perfectly accurate.

There are four general problems one typically encounters with frames: eligible population members not on the list, ineligible members included in it, some eligibles appearing multiple times (duplication), and the reverse: some group of two or more eligibles having only one representation on the list. The first two problems are of concern because they create a mismatch (i.e., undercoverage or overcoverage) between the population of interest and the frame population. If those people who are omitted in error differ (in terms of the study variables) from those included, it clearly represents a source of error in the resulting sample estimates. Inadvertently including nonpopulation members in the sample means that the sample estimates do not fully correspond to the target population.

The next two problems concern effects on the chances of selection. Selection is typically made from entries on the list (i.e., the frame elements). If some population members appear multiple times in the frame, they have a greater likelihood of selection than those who appear only once. Similarly, individuals in a group of population members combined into one listing have a lower chance of selection than the population members who each have his or her own separate listing. There are procedures for dealing with each of these problems. The point is that failure to deal with them has consequences, sometimes quite serious, for the quality of the sample estimates.

3.4 Sampling Error

One source of variation in a sample estimate is due solely to the fact that a sample, not the entire population, is selected. This source of variation is called the *sampling error*. In the process we have been describing, a sample (n) selected from a population and interviewed producing a set of responses (R) from the n respondents (assuming for the moment perfect survey cooperation). From this set R , the corresponding set of sample estimates (E) of the population values is computed.

There are many samples of size n that *could have been selected* from the population. Only one of those samples, n_1 , was actually chosen; and its set of responses, R_1 , gave us one set of estimates E_1 of the population parameters.

Suppose another sample (of the same size) is independently selected from this same population using exactly the same sample design, sampling frame, questionnaire, and data collection techniques (and that each survey obtains perfect cooperation). Now we have sample n_2 , the

set of responses, R_2 , and a second set of sample estimates, E_2 . Because everything else in the process is unchanged, any difference between the sets E_1 and E_2 is due solely to the fact that each collected data from a somewhat different set of respondents.

Sampling error is defined (conceptually) as the average variation on a particular statistic between all possible samples of size n selected from the same population in the same manner. Sampling error depends on (i.e., is a function of) the sample size, the sample design, and the amount of variation in the target population on the variable being measured.

First consider the sample size: With large sample sizes one expects less variation from one to the other. There are simply fewer possible large samples that can be selected from the fixed population. In addition, the sampling error depends on the sample design and is computed differently for different designs. A common error is to compute the sampling error without taking the design into account; another is to run statistical tests that assume a simple random sample (discussed below) without taking into account that the survey sample was selected using another design. Finally, regardless of sample size or design, variation exists in the sample simply because individuals in the population differ from one another. Although this is not technically a sampling error (it is a reflection of true population variation), it is included in standard measures of the sample error.

In summary, the general sampling procedure outlined so far requires that the researcher define the target population, obtain a suitable sampling, determine the necessary sample size, select a probability sample, and compute sample estimates of population parameters. The next sections outline both simple and more complex sample designs.

4 Available Sample Designs

4.1 Simple Random Sampling

The objective of many, but by no means all, sample designs is to give every population member the same chance of selection as every other population member. One sample design that accomplishes this is simple random sampling.

With simple random sampling, each member of the sampling frame is numbered 1 through N (i.e., $1, 2, 3, \dots, N$), where N is the total number of listings. If a sample of size n is desired, then n random numbers are selected (in the range 1 to N) from a table of random numbers or a random number generator. Each random number that corresponds to a number $1-N$ identifies a selected sample member (numbers selected more than once are ignored after their first occurrence). There are more details, but this gives the general conception. A simple random sample gives each member of the sampling frame the same chance of selection, and that chance is n/N . Moreover, with simple random sampling *every possible sample* of size n has the same chance of being selected. This implies that every member of the population, N , has the same chance of selection, which is what we are usually interested in.

Simple random sampling has several practical shortcomings. We focus here on just two. First, it can be a rather cumbersome procedure to implement partly because most lists used for surveys were not created for sampling purposes. Unless the sampling operation is computerized, numbering a list is time-consuming and tedious. If the sample is even moderately large, many random numbers must be selected (generated) and duplicates recorded. Second, although unlikely, it is possible that a simple random sample now and then produces a sample with a distribution quite unlike that of the population from which it was selected. This is more likely when the sample is small. Such “odd” distributions rarely occur, but it would be good to avoid them altogether. Depending on the order of the list, systematic sampling can reduce the chances of such distributions; stratified sampling can eliminate them altogether.

4.2 Systematic Random Sampling

Simple random sampling gives each possible sample the same chance of selection; this is more than we require for most survey sampling purposes. It is the individual’s chance of inclusion that concerns us; this is a goal that can be attained in a simpler fashion. Systematic random sampling is a more common method used to sample a list by selecting every k^{th} person on the list (sampling interval). This is clearly a simpler procedure, especially when the population list is large. This technique differs from simple random sampling in that every possible sample of size n no longer has a chance of being selected (e.g., using this method there is no way that two population members next to each other on the list can be in the same sample). The number of possible samples has been greatly reduced. In fact, the number of possible samples is exactly equal to the sampling interval, but each individual list member has an equal chance of inclusion, and that is the primary goal.

In a sense, this approach will produce perfectly acceptable samples that meet the requirement of “representativeness” as defined earlier. However, many times there are practical obstacles to using this method; for example, it may be that no satisfactory single frame is available. More importantly, however, there are often ways we can improve the effectiveness and efficiency of the design. By *effectiveness* we mean the precision of the sample estimates for a given sample size, and by *efficiency* we mean the cost of implementing the sample in an actual survey. These are important issues when sampling easily enumerable populations; they become critical when the population is rare or elusive.

4.3 Stratified Sampling

Assume that a population contains sets of individuals similar to each other on some variable, such as education, neighborhood of residence, or some other demographic characteristic. If that variable is also related to the survey’s substantive measures, such as health-related behavior or leisure activities, it may be possible to improve the sample design efficiency by grouping similar individuals together into *strata* and then

sampling each stratum independently. This assumes that information about the stratification variable is available in the sampling frame. (There are methods for stratification that can sometimes be used when the variable is not in the sampling frame, but this discussion does not address those complications.)

There are two general types of stratification: proportional and disproportional. Both begin with defining the strata of interest and then follow with dividing the population/frame into strata and selecting a portion of the total sample from each stratum. With proportional stratification, the percentage of the sample allocated to each stratum equals that stratum's proportion of the total target population. With disproportional stratification, some (perhaps all) strata receive sample allocations that are either more or less than their percentage of the total target population.

4.3.1 Proportional Stratification

Proportional allocation of the sample is used to improve the precision of the sample's representativeness of certain important characteristics. For example, assume one was to sample female clinic patients from a list that contained each woman's age, and age was thought to be correlated with some outcome variable. Before sampling, the list could be rearranged into age strata: 18 to 29, 30 to 39, 40 to 49, 50 and older. A sample would be selected from each age stratum proportional to that stratum's percentage of the total female patient population. If no stratification was done, these are still the proportions that would be expected on average over repeated independent samples. However, any one particular simple random sample might, by chance, underrepresent or overrepresent a particular stratum, even by a large amount. If the stratum variable is related to the dependent variables, it would be good to avoid the possibility of such a sample. This is what proportional stratification does. It is not a very powerful technique, meaning that it usually produces only modest improvements in the precision of sample estimates. If proportional allocation can be done inexpensively (e.g., by reordering a computerized list before sampling from it), it can be useful. It is generally not worth a large investment of study resources.

4.3.2 Disproportional Stratification

Disproportional stratification is a much more versatile and powerful design. With disproportional stratification, the proportion of the sample allocated to a stratum is not the same as that stratum's proportion of the population. There are several broad reasons to use disproportional stratification that are relevant to and summarized in this chapter.

The first general reason for using disproportional stratification is because the strata are of interest in themselves. In many surveys, in addition to estimates for the total target population, the researchers may be interested in either separate analysis of certain subgroups or comparing some subgroups to others. Often the natural proportions do not produce enough cases for these objectives.

Assume that in a survey of lesbians the main objective is to compare lesbians who live in the inner city of a metropolitan area to those who live in the suburbs. The most efficient sample allocation *for this objective* is to take half the sample from the city and half from the suburbs, even though the actual distribution of the total lesbian population may differ considerably from a 50:50 sampling distribution. This allocation minimizes the sampling error of differences found between the two strata, the main type of analysis for a study comparing subgroups (strata).

Another common objective is to ensure that enough of a population subgroup is sampled to permit separate analysis. Assume, in the same survey of lesbians, that 80% of the target population are expected to reside in the suburbs and 20% in the inner city. If the total sample were 400 interviews, only about 80 would be expected to fall in the city if no stratification was used. If 80 were considered too few for the planned analysis, one might, for example, take a double sample from the city to produce 160 interviews. It is useful to keep in mind that even within a targeted subgroup further subgroups may be of interest. So, even if 80 cases were considered marginally acceptable for examining inner-city lesbians in total, if further analysis by age or education was also planned at that level the 80 cases would likely be insufficient. Once again, it is seen how the sample design depends on the planned analysis.

A second reason to use disproportional stratification is that costs differ by stratum (i.e., certain sample members may be more expensive to survey than others). When surveying a population that is not evenly distributed geographically (e.g., Hispanic adults), it may be much more costly to locate sample members in some areas (in this case, the northwestern United States) than in others (e.g., the southwestern United States).

Sample designs for rare populations may have widely varying costs for the same reason (i.e., the distribution of the population). Even small absolute differences can have major effects. Consider sampling for a population that has 5% prevalence in one stratum and 12% prevalence in a second stratum. Although the difference is only 7%, more than twice as much screening is needed in the first stratum than in the second one to locate each sample member.

The usefulness of disproportionate stratified sampling for rare groups depends on the extent to which the prevalence of the target group varies across strata (e.g., geographic areas). For the procedure to be effective, some geographic areas need to have *both* a higher prevalence of the target group *and* include a large proportion of the total target population. This may be best illustrated with a simple example.

Assume that for a citywide, household survey of gay males three strata are identified. Stratum one consists of a set of neighborhoods known to have a high prevalence of gay males, say 60% of all households. Stratum two is a set of more "mixed" neighborhoods also known to have a high prevalence of, say, 15% of households. Stratum three is the remainder of the city, where the prevalence is expected to be 3%. Clearly, the cost per case, which depends largely on the screening

costs to locate eligible households, varies greatly from one stratum to another. A sensible sample design oversamples *to some extent* strata one and two. Consider two possible situations, keeping in mind that people who reside in certain neighborhoods may very well differ both demographically (e.g., age, income, education) and in terms of the substantive variables being measured from those who choose to reside outside those neighborhoods.

In the first situation, assume that *of the total gay male population of the city* 50% reside in stratum one, 25% in stratum two, and the remaining 25% in stratum three, the rest of the city. This population distribution is fortunate for sampling purposes. The strata that have high prevalence rates (percentages of stratum households that are eligible) also contain a large proportion (75%) of the target population. So it is statistically logical to heavily oversample those strata.

Now consider a second situation in which stratum one contains 15% of all the city's gay males, stratum two has an additional 15%, and the remaining 70% are in stratum three, which happens to have the lowest within-stratum prevalence. Although strata one and two can still be oversampled to some extent, the sample cannot be heavily concentrated there because even though the density of eligibles is high *most of the city's gay males do not reside in those neighborhoods*. Although the technical details of computing the correct sample allocations is beyond the scope of this chapter (see Kalton, 1993, for those procedures), the reasoning behind disproportional stratification is that one has to consider each stratum's prevalence *and* the percentage of the entire population it includes to design an efficient sample. This is analogous to a national survey of Hispanics in which it is known that New Mexico has a high prevalence of Hispanics (i.e., most of New Mexico's population is Hispanic), but New Mexico contains a very small proportion of all the nation's Hispanics. It would be unwise to concentrate a national sample in New Mexico. It would be cost-effective, but the final sample would not accurately reflect the nation's Hispanic population.

Finally, in regard to costs differing by strata, if multiple methods of data collection are used, such as telephone and face-to-face interviews (for those without telephones), the face-to-face sample may be considered a separate stratum for which the cost per interview is greatly increased. In each of these cases, an allocation of the sample that is disproportional to the population distribution may be necessary.

When each sample member has the same probability of selection, we describe the sample design as "self weighting." Each sample member represents the same number of population members; put another way, each population member is equally represented. When (and this can come about in various ways) sample members have different probabilities of selection, some population members are overrepresented relative to others. This needs to be accounted for to produce unbiased sample estimates of population parameters.

Recall an earlier example. In a disproportionally stratified sample of lesbians, those living in the city were oversampled relative to those in the suburbs. In one instance, this was to compare the two groups optimally; in another, it was to have sufficient cases for separate analysis.

Both uses of stratification were justified for those analytic purposes. Usually, however, one would also want to combine the two samples for total target population estimates (i.e., for the whole metro area). If city and suburban residents differed on substantive variables, the city residents would be overrepresented and the resulting sample estimates biased in their direction. To produce unbiased total estimates, these different probabilities of selection (for those particular analyses) must be corrected. This is accomplished with weighting. The details of weight construction are beyond the scope of this discussion, but the effect is that, in this example, suburban residents are assigned a greater weight to make up for the fact that there are fewer of them in the sample than their portion of the population requires.

With weighting, each sample member's response to a substantive question is multiplied by that sample member's "weight." The *weight* is a number that can be greater than 1, meaning that the response is counted more heavily in the estimate (as if there were more of that "type" of sample member in the sample). This type of weight makes up for underrepresentation (having a lower probability of selection than equal representation would require). Similarly, a case can have a weight of less than 1, achieving the opposite effect of reducing the impact of that case's answer on the estimate.

4.4 Cluster Sampling

With the sample designs discussed to this point, respondents have been selected essentially one by one. There are many instances where considerable cost savings can be realized by selecting groups or clusters of respondents. Clusters are *naturally occurring* groups of potential respondents. Often costs can be reduced by sampling clusters, but there is a statistical *cost*. There are often similarities between people in a natural cluster. That is, people who live on the same city block, those treated at the same clinic, or those who are in the same school classroom may be alike in some way, perhaps by age or income. People are not grouped into clusters at random. Put another way, the interview measures within a cluster may be correlated; that is, they are not completely independent observations. This means that a sample of size n selected by way of clusters is almost always inferior to a sample of the same size selected by a simple (or systematic) random sample; that is, the sampling error is larger for the same size sample).

Recall that the sampling error is partly a function of sample size; thus, as sample size increases, sampling error decreases. The way around the problem of within-cluster homogeneity is to increase the total sample size to offset its effects. The cost per case is often so much lower with cluster sampling that, for the fixed cost that constrains real-world surveys, a larger sample can be selected to offset the effects of within-cluster correlation.

Although the main rationale for cluster sampling is lower sampling errors for the available fixed costs, there are other reasons as well. A common reason is the unavailability of sampling frames for many populations.

An important application of cluster sampling for rare populations is telephone cluster sampling (TCS). TCS is a variant of Mitofsky-Waksberg (Waksberg, 1978, 1983) sampling applicable to rare groups and was described by Blair and Czaja (1982). With TCS, a random telephone number is dialed in a bank of telephone numbers (a bank is 100 numbers generated by randomizing the last two digits of telephone numbers). This number can be selected via list-assisted random digit dialing or any other procedure.¹ If the number is found to be a working household number, the household (or person) is screened for membership in the target group. If the household is not a member of the target group or if the number is not a working household number, no further sampling is done within that bank. However, if a group member is found, further sampling is done within the bank until a pre-specified number of group members are identified.² This procedure has the effect of rapidly dropping telephone banks with no target group members.

The usefulness of TCS for sampling a rare group generally depends on the extent to which the group is geographically clustered. If the group is spread evenly across telephone exchanges, and there are few phone banks in which it does not occur, then TCS increases the operational complexity of the research without improving its efficiency.

An alternative perspective to geographic clustering for deciding whether TCS will be efficient is to consider the effectiveness of TCS using ρ , the intraclass coefficient of homogeneity, as an indicator of the rare group's tendency to cluster within telephone banks (ρ is the conventional measure of tendency for similar elements to co-occur within clusters (i.e., the correlation between elements within clusters with regard to the characteristic of interest) (cf. Sudman, 1976)).

Assume we use the TCS method described earlier, in which subsequent calls in a telephone bank are made if and only if the first call produces a member of the rare group. Let π be the proportion of the general population who fall into the target group and thus the expected screening rate for "first calls" into random telephone banks (ignoring, for the moment, the effects of nonworking numbers, business numbers, noncontacts, and refusals to participate). Let π' represent a comparable screening rate for second and subsequent calls into telephone banks

¹The original purpose for Waksberg sampling was to find working household telephone numbers and eliminate banks of nonworking numbers. For this purpose, list-assisted RDD competes with Waksberg sampling (list-assisted methods eliminate nonworking banks by restricting the sampling to banks that are known to have at least one listed household). However, when the goal is to find members of a rare group, the two procedures are complementary. List-assisted methods can eliminate banks of completely nonworking numbers, and Waksberg sampling can eliminate working banks in which the target group does not occur.

²The cluster size is defined as identified, not cooperating, eligible households or individuals. If the cluster size is k , calling in the bank stops after k group members are identified by screening, regardless of whether they consent to the main interview.

where the first call has produced a member of the rare group; in other words, π' represents the conditional probability of drawing a member of the group given that one has already been drawn. Finally, assume that ρ is the intratelephone bank coefficient of homogeneity for the defining characteristic(s) of the target group, defined as the correlation in incidence between successive elements within banks (e.g., “first call” and “second call”). It can be shown (Blair & Blair, 2004) that the relationship between π and π' is as follows:

$$\pi' = \pi + \rho(1 - \pi)$$

Essentially, this expression indicates that if the first number called in a bank is eligible, the prevalence of eligibles in that bank π' is greater than the overall eligibility rate π . The actual screening rates experienced in a research project are lower than indicated because of nonworking numbers, business numbers, noncontacts, and refusals.

The details of the analysis of the efficiency of TCS are beyond the scope of this chapter, but the conclusion of that analysis is that the rareness of the target group was found to be at least as important as the level of geographic clustering (and arguably more important) for determining the relative efficiency of TCS. The intuition is that as first-stage respondents become more difficult to find it becomes increasingly beneficial to look near those respondents for others like them if there is even a mild tendency for respondents to cluster. On the other hand, if first-stage respondents are easy to find, there is little to gain from clustered sampling, even given a strong tendency to cluster.

This design has been incorporated into an “adaptive sampling” approach to produce an efficient two-phase approach for locating a population of men who have sex with men (MSM) in a multicity survey (Blair, 1999).

5 Complex Sample Designs

5.1 Adaptive Designs

With all the designs described to this point, the sample design is “fixed.” That is, using the best information available at the outset of the study, a sample design is selected, sample size determined, sample allocated to strata, and so on; and the survey proceeds. In short, the sample is selected and its procedures followed to select and interview sample members. Another class of sample designs are termed adaptive samples. “Adaptive sampling refers to sampling designs in which the procedures for selecting sites or units to be included in the sample may depend on values of the variable of interest observed during the survey” (Thompson & Collins, 2002). This is an appealing approach for some LGB surveys, although it is easy to underestimate the complications involved in its careful implementation.

Sample designs rely to varying degrees on prior information about the target population. In the case of rare or elusive populations, the estimate of prevalence of the target population within the larger pop-

ulation is one such piece of information that is especially important—even more so if the prevalence varies significantly by geography or other sample units (as was the case with TCS). In some situations this prior information about prevalence may be suspect for various reasons. The prevalence information may not exactly match the survey's population definition, or it may be dated. Different sources of information may produce different prevalence estimates. The data source may be from the census, from some type of administrative records, or from a survey that did not involve screening for the particular population. In many such instances the degree of underreporting may differ markedly from what would be found in a screening survey. Finally, data from multiple sources may be combined into one "best value" estimate of prevalence. The procedure for combining the data has error properties that may be difficult to specify. All these possibilities produce situations in which crucial design decisions are based on imperfect data.

Adaptive sample designs may permit a quantitative assessment of the prevalence data before all the sampling is completed. Consider, for example, a stratification design in which costs differ by stratum. As noted earlier, disproportionate stratification may be an efficient design when costs differ substantially by stratum. Following Hansen et al. (1953), Sudman (1976) produced a simplified formula relating the cost per interview to the sample allocation to strata. Essentially, a proportionally larger percentage of the sample is allocated to strata in which the cost per case is lower and less of the sample to the strata with higher cost per case. When the target population's prevalence is the only factor that differs by stratum, the cost per interview is directly proportional to the prevalence. The estimated cost per interview, and hence the strata allocations, depend on knowing the target population prevalence in each stratum.

An adaptive design using disproportional stratification owing to differential costs would release a small portion of the sample, based on the estimated prevalence, in each stratum for surveying. The screening rate for each stratum would be determined based on contacting the released sample. That is, the estimate of prevalence would be corrected based on actual screening. To the extent that the screening rates differ from the expectations, the remaining (unreleased) sample would be allocated based on this new information. Even though the screening rates are themselves estimates, subject to sampling error, it is possible to detect major differences between the expected and actual prevalence. This approach was used to improve the sampling efficiency of a survey of gay urban males in multiple citywide surveys (Blair, 1999).

5.2 Network Sampling

In a survey to locate a target population with a particular characteristic, the most direct procedure is, as we have seen, to screen a random sample of households and for each one determine whether someone with the characteristic of interest resides there. If the target population is rare, this procedure requires screening many households. The

reporting rule, in such a conventional survey, asks a household to report about all of its members and only those members.

Network sampling uses a different reporting rule. Households in the selected sample are conceptually linked to other households (which almost always are not in the sample). Some of those other households contain a member of the target population. The reporting rule asks a sample household to report about its own residents *and* about the residents of households to which it is linked. The screening rate is increased to the extent that target population households are linked to households in the sample.

Birnbaum and Sirken (1965) first described this sampling approach as a method for estimating the prevalence of rare diseases. The method has been used or tested as a technique to identify cancer patients (Czaja et al., 1986), crime victims (Czaja & Blair, 1990), and other rare populations (Sudman et al., 1988).

There are various possible applications of network sampling to LGB surveys when the objective is to identify these target groups within the general population. The first and intuitively most appealing application is to use network sampling to identify and interview target population members. With this approach, respondents are asked whether the members of some prespecified social network (e.g., their brothers and sisters) have the characteristic(s) of interest. If any member of the network belongs to the group, the respondent is asked for contact information, and the researcher attempts to interview those network members.

Network sampling is effective only if the defining characteristic of the group, in this case sexual orientation, is known to other members of a measurable network. Also, if network members are to be interviewed, the initial respondents must be, in the main, willing to provide referrals to other network members. Even then, the benefits of the procedure are offset by the difficulty of having to find networked respondents, the difficulty of developing reliable weights to correct the differential inclusion probabilities resulting from varying numbers of people who might have nominated them, and design effects from weighting. Even if all these conditions are met, if the researcher's target population is limited to a specified geographic area, then identified network members who reside outside that area are ineligible for the survey. Because of these limitations and difficulties, network sampling is rarely used to locate and interview target population members.

There are two other contexts where the procedure might be effective. First, when the purpose of the research, or some phase of it, is to estimate the prevalence of the target group rather than contacting its members, the researchers might choose network sampling. If all one needs is prevalence data, then network sampling has the potential to expand the effective sample size without imposing the difficulty of obtaining referrals and finding networked respondents. This, in fact, is how the procedure was used in its early days (cf. Sirken, 1970), but that application has faded.

The second possible application is related to the first. Network sampling might prove useful when adaptive procedures are employed to

guide disproportionate stratified sampling, and prevalence data are needed to guide the strata definitions and allocations. Here, one is in the position of estimating prevalence based on relatively few data in any given geographic area, and the increase in effective sample size associated with network sampling might be helpful.

In both contexts, it is still necessary that group membership is identifiable by other network members, and it is still necessary to estimate or control the number of people who might nominate any given network member. Also, in the latter context, because network measures are used to inform a geographically based stratification scheme, it must be possible to link network identifications to that scheme (i.e., determine in which stratum a nominee lives). This might be done, for example, by obtaining the ZIP code or the first four digits of each network member's telephone number. These requirements continue to limit the use of network sampling; but considering the value of prevalence information in studies of rare groups, the possibility of using network sampling should not be overlooked.

5.3 Site or Time/Location Sampling

With site sampling, visitors to a well defined location are sampled at that location. The sample may be selected at one point in time, multiple discrete points in time, or over some continuous period of days or weeks. Locations can be single retail establishments, groups of such establishments (e.g., a retail chain or a mall), and recreational locations such as parks or theaters; even temporary settings such as street fairs can, under certain conditions, be sampled. Sampled respondents usually answer questions about themselves but can be asked about the particular establishment instead of, or in addition to, personal information.

Site or time/location sampling, properly executed, is a probability design applied to a limited, nonhousehold population. Site sample designs are used when the particular site or location is of interest in itself, or when the site is a gathering place for a population of interest. The population may be sampled on site because the site is part of the population definition (e.g., patrons of lesbian bars) or simply because the site is convenient to find members of a more broadly defined population (e.g., all lesbians in a particular geographic area). All visitors to the site may be eligible sample members or only those with particular characteristics.

Although such samples may be appealing for a number of reasons, two cautions should be noted. First, typically the population of inference is severely restricted, often representing only visitors to the particular location(s). Second, if inappropriate sampling procedures are used, the probability design can quickly degenerate into a sample of convenience.

A limited definition may or may not be useful. If for example, the site is one gay bar that happens to be accessible, for whatever reason, to the researcher, the sample represents only patrons of that bar. It does not represent all bar patrons and certainly not all gays in a city or

neighborhood. For some purposes, that definition boundary might be acceptable.

It is possible to expand the scope of the definition by selecting a probability sample of establishments and then sampling patrons within each one. For example, if a list of all gay bars (assuming that "gay bar" can be rigorously defined) in a town is available, a sample of bars could be selected using probability sampling, and site sampling could be done at the sampled bars. The population definition is thus expanded to cover patrons to bars (of the selected type) in that town.

The bars could either be chosen with equal probabilities and the sample allocated across them in proportion to an estimate of each bar's size (total patrons per week or some other measure related to the size of the target population the bar's patrons includes). If the entire census of bars were included, proportional allocation of the sample would also be the simplest legitimate approach. Alternatively, a sample of bars could be selected with probabilities proportional to size. In that case, a fixed number (cluster size) of sample members would be selected at each bar. The latter approach would be useful for practical reasons (i.e., to equalize the burden on each bar); or the same allocation might be an efficient way to ensure enough sample of all size (or other characteristic of) bars for subgroup analysis.

Assume a site has been selected by whatever means: How does site sampling work? Most often, a two-stage, cluster sample design is used. In the first stage, time slots are selected; in the second stage, respondents visiting the establishment during those time slots are sampled. If the time slots are selected with probabilities proportional to size (where the measure of size is number of expected visitors or some correlate of that, such as sales), then a fixed cluster size of respondents is chosen. The two stages taken together produce an equal probability sample of visits. It is important to note that the sample is of visits, not visitors. The probability of an individual visitor is proportional to the number of times he visits the establishment during the data collection period. The sample of visits may be weighted to convert it to a sample of individual visitors, as shown below.

The two-stage cluster approach was used to select a sample of patrons for an exit survey of patrons at a gay bathhouse. We are including this example because a time/location design is often used for sampling LGB populations, and detailed examples of how it is used are not easily available elsewhere, unlike examples of other, more conventional sample designs. The following description focuses on the estimation procedure, which is essentially a process of constructing weights to produce unbiased sample estimates. This detailed description also illustrates some general principles of sample weighting for non-response and for differential probabilities of selection. The general procedure is to construct individual weights, including (1) a "base weight" to account for the overall probability of selection (this base weight, which theoretically is a simple inverse of the probability of selection, has to be constructed taking into account some practical steps in sample selection); (2) a weight to convert the sample of visits into a sample of individuals, the units of analysis; and (3) a weight to adjust

for differential nonresponse. These individual weights are then combined into one overall weight for data analysis.

The participating bathhouse was open 24 hours a day, 7 days a week. A time probability sample was selected over a 5-week period. Using data on the number of customers per employee work shift, a probability sample of shifts was selected during week 1, with probabilities proportional to the number of customers expected (i.e., busy shifts were selected more often). Within each selected shift, a systematic sample of customers was selected with equal probabilities. During the subsequent weeks, the expected number of customers per shift was updated based on counts in the sampled shifts. Using this sampling design, we sampled 708 patrons exiting the bathhouse, with 440 (62.1%) agreeing to participate in the survey. Those declining to participate were not significantly different from participants by age ($p = 0.38$) or race ($p = 0.73$). There was a significant difference in refusals during the 3 to 6 PM shifts ($p < 0.003$), where we averaged 57% of the men refusing compared to all other shifts, which ranged as low as 23.5% (3 to 6 AM) to as high as 48.7% (9 AM to noon). There was also a significant improvement in acceptance as the weeks passed, with the highest rate of refusals occurring in week 1 (48.2%) and the lowest in week 5 (25.3%).

This time-probability sampling design did not produce a self-weighting sample of either visits or of individuals. A series of weights were necessary to produce unbiased, or nearly unbiased, estimates. Strictly speaking, each week, k , has a base weight

$$WB_k = 1/P_{vk}$$

the inverse of probability of selection in week k , where

$$P_{vk} = P_{sk} \times P_{(s)k}$$

Hence, the probability of a visit P_{vk} during a given week k is the probability of a shift being chosen

$$P_{sk} = N_{sk} / \sum N_k$$

times the probability of selection within the shift.

$$P_{(s)k} = n_{sk} / N_{sk}$$

The complication in computing this weight is that we have only an estimate of the total number of visits for a week:

$$\sum N_k$$

Counts of visits are available only for the shifts that were sampled. For the nonsampled shifts, the counts are either the previous weeks' counts or the initial count provided by the establishment.

To simplify this computation, we have assumed a fixed mean of 600 visits each week, for a total of 3000 visits for the 5-week period. The probability of selection of a visit is then modeled as the ratio of the total sample selected over this estimate of total visits.

The sample size for a shift was modeled as the target number of completed interviews, 6, adjusted for the mean expected response rate 50%, $6/0.5 = 12$. The selected sample varies when the actual number of visits for a shift differs from the expected measure of size for the shift. The probability of a visit over the entire data collection period is estimated as

$$P_v = \sum_k \hat{n}_k / \sum_k \hat{N}_k$$

where $\sum_k \hat{N}_k = 3000$ and

$$\sum_k \hat{n}_k = 12 \sum_i s$$

or 12 times the number of shifts selected over the 5 weeks.

5.3.1 Visits Weight

A respondent visits weight (WV) is necessary to adjust for differential numbers of visits by different respondents during the 5-week data collection period. This weight essentially changes the sample from a sample of visits to a sample of individuals.

The probability of person j being selected is proportional to that person's total visits during the data collection period (obtained from survey questions).

$$P_{vj} \propto \sum_i v_{ij}$$

The "visits" weight is the inverse of the estimated total visits for a person.

$$WV_j = 1 / \sum_i v_{ij}$$

5.3.2 Nonresponse Weight

The shift nonresponse weight is used to adjust for situations when the actual number of completed interviews for a shift (c_a) falls short of the target cluster size (c_t) (of completed interviews) for that sampled shift.

$$W_{nr} = c_t / c_a$$

This weight adjusts both for shortfalls due to nonresponse and due to an inadequate number of visits in the shift to reach the target c_t .

5.3.3 Total Weight

The total weight (WT) is obtained by combining all three weights into one weight variable.

$$WT = WB \times WV \times W_{nr}$$

This total weight (WT) then weights the sample back to the estimated total population.

5.4 Cautions When Using Complex Designs

These complex sample designs have been developed to handle the special problems of cost-effectively sampling rare or elusive populations. Some of these approaches are extensions of basic designs such as disproportional stratification; others, such as network sampling, differ radically from conventional sample designs.

Some points of caution are relevant here. Complex designs typically require more specialized expertise to select the sample and may also rely on difficult procedures being properly executed as part of the data collection. Even when the idea underlying a design is fairly simple, its implementation may not be. A second caution flows from the first. When a design is badly implemented, its advantages may be lost or may be attained at an unacceptable price, such as when cost savings are realized in return for major increases in sampling variance or high nonresponse rates. Such unfortunate consequences may be present, even if they are overlooked by the researcher. An even more serious consequence of poor execution is that the sample may cease to be a probability sample at all, such as when incorrect sampling procedures make it impossible to specify probabilities of selection or to produce valid sample estimates. Additionally, some of these designs are based on assumptions or conditions, such as that respondents can provide certain information or perform particular response tasks. Some of these required conditions can be easily met, whereas for others it is quite problematic, even dubious, that they can be realized. If a design rests on strong assumptions, they must be carefully assessed, perhaps by conducting a pilot study, before proceeding.

Finally, when using complex sample designs, it is important to note that although for many designs there are formulas to compute sampling errors directly, in many instances such formulas do not exist. It is standard practice today to use one of several software packages that have been made available in recent years for the correct computation of sampling errors for complex surveys. SUDAAN and WesVar are two of the most popular and statistically sound. A common error of novice researchers is to apply the formula for simple random sampling errors to other designs, leading to often severe underestimation of sampling errors. Even though a complex design may produce equal probabilities of selection for individual sample members after many steps and statistical weighting adjustments, it by no means constitutes a simple random sample.

6 Research of LGB Populations Using Probability Samples

As we have discussed in this chapter, probability sampling allows researchers interested in LGB topics to generalize their findings to a larger population (i.e., beyond the sample itself). This might involve estimating the size of a particular population or the prevalence of a specific phenomenon.

Approaches to probability sampling in LGB research have fallen into three broad categories. The first category includes designs where investigators draw a probability sample of the general population (gay and nongay) and then conduct analyses on the subsample of LGB respondents within the larger sample. In this case, the LGB individuals are a subpopulation of the larger population, and research findings from the subsample of LGB respondents can be generalized or projected back to that subpopulation. In the second category, researchers may employ techniques to increase the size of the sample with desired characteristics (e.g., homosexual identity or HIV infection). These techniques typically involve focusing on preselected ZIP codes, census tracts, or social settings thought to have a greater concentration of the desired characteristic and then drawing a probability sample of all or a segment of the population only from within that focused area. The primary benefit of such techniques is that the subsample of interest (e.g., LGB persons) is larger than what could be sampled in general population studies. However, such techniques also limit the ability of investigators to generalize to the LGB population beyond the selected ZIP code or setting. The third category of designs is similar to the second, except that investigators draw a probability sample by screening for *LGB individuals only*, again generally in selected ZIP codes or settings. In these cases, generalizations are made from the LGB probability sample to the LGB population in the selected ZIP code or setting and generally result in larger samples than the previous categories, which allow for more extensive analyses.

Table 15.1 lists several examples of studies in each of these categories of research. For each of the studies cited in Table 15.1, we list the population the researchers were interested in studying, the substantive research interest, the sample design, the actual population that was sampled, and the mode of data collection. Our goal was to cite studies that displayed a range of population interest, type of probability design, and mode of data collection. When possible, we selected studies that described the sample design in some detail to provide the reader with more information than is generally provided in journal articles. For some studies the same design is described in the substantive article or book chapter listed; for others additional articles or chapters that provide extensive discussion of the sample design are cited for more detailed descriptions.

Not surprisingly, most of the LGB-related studies using probability sample designs are quite recent. The funding that provides researchers with sufficient resources to draw probability samples of populations reporting same-gender sexuality is a recent phenomenon (last 15 years)

Table 15.1. Examples of Probability Sample Designs in Studies of Lesbian, Gay, Bisexual Adults, and Adolescents

Population of interest	Research interest	Sample design and [mode of data collection]	Population sampled	Reference
<i>Subpopulation of LGB in a general population</i>				
Lesbian, bisexual women	Patterns of cigarette smoking and alcohol use among lesbians and bisexual women	Stratified random sample from membership list [self-administered questionnaire (SAQ); mail]	Kaiser Permanente members residing in northern California region at least 20 years of age and members at least 3 months	Gruskin et al. (2001)
Men reporting same-sex sexual contact	Prevalence of men who have sex with men (MSM)	Multistage area probability samples from five samples [personal, SAQ]	U.S. adult population (ages varied by sample); also sampled residents of Dallas County, Texas aged 18–54; U.S. adult population	Rogers & Turner (1991)
Women and men reporting same-gender sex, desire, attraction	Comparison of men/women same-gender sexuality	Multistage area probability using multiple samples [personal, SAQ]	contiguous states (ages 18–59)	Michaels (1994)
Men reporting same-sex sexual contact	Prevalence of MSM nationally, and in urban, suburban, and rural areas of the USA	Multistage area probability sample and random digit dialing (RDD) sample [personal, SAQ, phone]	U.S. adult population (ages 18–49)	Description of methods: Laumann et al. (1994, Chapter 2 and Appendix A) Binson et al. (1995)
Women and men reporting same gender sexuality	Prevalence of homosexual behavior/attraction in USA, UK, and France	Multistage area probability sample [personal/SAQ]	Adult populations of USA, UK, and France (ages 16–50)	Sell et al. (1995)
Same-sex and opposite-sex cohabiting couples	Comparison of partner selection among same-sex and opposite-sex couples	Multistage design [SAQ]	Cohabiting couples/households (ages 19–65) from 5% sample of 1990 Public Use Microdata Set (PUMS) of 1990 U.S. census	Jepsen & Jepsen (2002)
Black same-sex cohabiting couples	Comparison of demographic characteristics of Black and White same-sex and opposite-sex couples	Multistage design [SAQ]	Cohabiting couples/households from the 5% PUMS of 2000 U.S. census	Dang & Frazer (2004)

Table 15.1. Continued

Population of interest	Research interest	Sample design and [mode of data collection]	Population sampled	Reference
Adults with same-sex sexual partners	Prevalence of psychiatric syndromes	Multistage probability sample [personal]	U.S. noninstitutionalized adult population (surveyed through 1996 National Household Survey of Drug Abuse)	Cochran & Mays (2000)
Gay, lesbian, and bisexual (GLB) adolescents	Relationship between gay-sensitive HIV instruction and risk behaviors of GLB youth	Multistage probability sample of schools, classes [SAQ]	Students enrolled in grades 9–12 in public high schools in Massachusetts	Blake et al. (2001)
Men reporting same-sex sexual orientation	Prevalence of homosexual and bisexual men	RDD sample [phone]	U.S. adult population from all 50 states from ABC-Washington Post poll	Harry (1990)
Youth with same-sex attractions	Relationship between same-sex attraction and experiences of violence	Multistage sample of USA feeder and high schools [Audio Computer assisted self-interview (ACASI)]	Students enrolled in grades 7–12 in sampled high schools and feeder schools in USA	Russell et al. (2001)
<i>Subpopulation of LGB in selected settings or ZIP codes</i>				
Homosexual, bisexual men	Prevalence of HIV-related risk and HIV status	Multistage area probability sample [personal]	Single men (ages 25–54) residing in 19 selected census tracts of San Francisco	Winkelstein et al. (1987)
Young homosexual and bisexual men	Prevalence of HIV status	Multistage area probability sample [personal]	Single men (ages 18–29) residing in 21 census tracts in San Francisco	Osmond et al. (1994)
Lesbian and bisexual women	Feasibility of assessing sexual orientation among an RDD sample of women	RDD sample [phone]	Women (ages 18–59) residing in Boston's Jamaica Plain neighborhood	Meyer et al. (2002)

Table 15.1. *Continued*

Population of interest	Research interest	Sample design and [mode of data collection]	Population sampled	Reference
Adolescent and young adult MSM in USA	Prevalence of HIV status and HIV-related risk behavior	Time/location probability design (multiple sites) [personal]	Men (ages 15–22) in public venues frequented by young MSM (e.g., urban shopping blocks, dance clubs, bars, social organizations, bathhouses, parks, beaches) in seven U.S. cities	Valleroy et al. (2000) MacKellar et al. (1996)
<i>Population of LGB in selected settings or ZIP codes</i>				
Homosexual, bisexual men	Prevalence of HIV-related risk	RDD sample [phone]	Adult MSM residing in 24 ZIP code areas of Los Angeles County (ages 18–75)	Kanouse et al. (1991)
MSM	Characterize AIDS epidemic among urban MSM	RDD sample [phone]	Adult MSM residing in selected ZIP code areas in four urban centers (ages 18 and older)	Catania et al. (2001) Description of sample design: Binson et al. (1996); Blair (1999)
Self-identified gay and bisexual Latinos men	Impact of discrimination and poverty on mental health	Time/location probability design (multiple sites) [personal]	Men entering bars, clubs, and weekend events primarily attended by Latinos and gay men in three U.S. cities	Diaz et al. (2001)
Young African American MSM	Social determinants of youth accessing HIV testing	Time/location probability design (multiple sites) [personal]	African American Youth (ages 16–25) entering venues frequented by young MSM (clubs, coffee houses, public parks, gay-pride events) in Atlanta, GA; Birmingham, AL; and Chicago, IL	Mashburn et al. (2004)

HIV, human immunodeficiency virus.

and, in the context of the AIDS epidemic, has been confined mainly to gay men. General population studies conducted by government agencies have only recently included questions related to same-gender sexual behavior. These data have provided opportunities for researchers to conduct analyses of health issues of populations reporting same-gender sexuality, particularly lesbian and bisexual women and adolescent populations.

The first and fourth columns of Table 15.1 describe the “population of interest” and “population sampled” of each of the studies. The “population of interest” is the group the researchers are interested in describing in some way (i.e., the group to which they want to generalize the results of their study). The “population sampled” provides the reader with information as to what population had some possibility or probability of being included in the sample (including geographic boundaries and age eligibility) and is an important factor to consider when thinking about who the sample represents. For example, in the Gruskin et al. (2001) study, the sample was Kaiser Permanente members residing in northern California who were at least 20 years of age. Hence, the population of lesbian and bisexual women to which one can generalize are those who have Kaiser Permanente health care, are residing in northern California, and are at least 20 years of age. Other examples of limited generalizability can be noted in the Centers for Disease Control and Prevention (CDC) study of young gay men (Valleroy et al., 2000) and the Urban Men’s Health Study (UMHS) (Catania et al., 2001). The sample frames for these studies are quite different, and it is important to recognize that the sample represents a particular stratum of men. In the CDC study the sample frame was public venues frequented by young men who have sex with men (MSM), and in the UMHS it was telephone numbers assigned to households in selected ZIP codes. In the case of the CDC study the sample reflected men who are comfortable going to public venues that cater to young gay men, and in the UMHS the sample reflects men who are comfortable residing in city neighborhoods that are likely to have high concentrations of gay men. In both studies (and in most studies of LGB populations), resulting samples tend to be more sensitive to populations who are comfortable identifying as gay or bisexual or frequenting gay venues (as in the CDC study) or living in gay neighborhoods (as in UMHS). The seriousness of such biases is unknown. One can only speculate about whether the differences between MSM, for example, who live in areas of the city with high concentrations of gay men and those who live in other neighborhoods are significant. One important concern is the research question being posed (i.e., how the data are to be used). Nevertheless, in all cases, the potential bias must be acknowledged and taken into consideration when reporting findings.

The focus of most of these studies is on individuals residing in the United States, although one study compares populations in the United States, the United Kingdom, and France. Two of the studies focus on couples rather than individuals.

6.1 Subpopulation of LGB Within a General Population

The first group of studies in Table 15.1 represents sampling strategies in which the general population is sampled (both heterosexual and LGB), and analyses are conducted on the subsample of LGB respondents. Random sampling from a list is a simple yet elegant design that allows the researcher to generalize the results to the larger population that was contained on the list. It is a relatively easy sample design if a suitable list is available, yet the Gruskin et al. (2001) study was the only one we could locate that used a list sample, in this case a membership list of Kaiser Permanente members in northern California. The research goal was to determine lesbian and bisexual women's cigarette smoking and alcohol use patterns.

In general, the population from which a LGB subpopulation is analyzed is geographically bounded, sometimes including the entire country or sometimes a smaller area, such as a state, a city, or even just a neighborhood. Studies that project back to subgroups or LGB subpopulations of larger populations involve drawing large sample sizes. One research interest is to analyze only the LGB portion of the sample. The Rogers and Turner (1991), Michaels (1994), and Binson et al. (1995) studies used various types of complex sample designs (multistage area probability design and random digit dialing) and various modes of data collection (personal, self-administered questionnaire, phone) to determine the prevalence of MSM in the United States and in large cities, suburbs, and rural counties. There is an excellent discussion in Laumann et al. (1994) of the multistage area probability sample design and the sampling procedures used for drawing the national sample for the National Health and Social Life Study (NHSLS) that was used in Michaels' (1994) work on the prevalence of same-sex attraction, desire, and sexual behavior among adult women and men in the United States. A similar sample design was used in Sell et al. (1995) in their study to determine the prevalence of same-gender attraction and sexual behavior in the United States, the United Kingdom, and France.

Several additional studies in this section are examples of innovative use of preexisting data from large probability samples to study LGB adults and adolescents. Only during the last 10 to 15 years have government and even private (e.g., Gallup or Harris) agencies included questions on their surveys that would allow researchers to select subsamples of LGB populations. The studies of same-sex couples (Jepsen & Jepsen, 2002; Dang & Franzer, 2004) used data collected as part of the U.S. census. Beginning with the 1990 census, individuals were allowed to select "unmarried partner" (bypassing the choice of "roommate") for the category "relationship to head of household." The Harry (1990) study used data from the ABC-Washington Post Poll to examine the prevalence of men reporting same-sex sexual orientation in the United States. Determining the prevalence of homosexuality is not particularly sensitive to the sample size (drawn from the general population) because the intended purpose is to determine the proportion of same-sex sexual orientation among sampled respondents. Cochran has

published a number of papers examining the prevalence of various psychiatric disorders among LGB adults (e.g., Cochran & Mays, 2000) using existing probability samples collected in large national studies of health (e.g., the National Household Survey of Drug Abuse, the National Survey of Midlife Development). In the two studies of lesbian, gay, and bisexual adolescents listed in Table 15.1 (Blake et al., 2001; Russell et al., 2001), respondents were sampled from in-school youth using a multistage design of first selecting schools and then classes in the selected schools. For some types of research, the small samples of LGB adolescents in the school samples can limit more complex analysis.

6.2 Subpopulation of LGB Within Selected Settings or ZIP Codes

The next group of studies in Table 15.1 are examples of sampling in geographic areas or in particular social settings in which LGB populations reside or socialize. In all four of the studies in this section, selecting census tracts, neighborhoods, or social settings based on the likelihood of high concentrations of the population of interest allowed investigators to recruit samples that included sufficient LGB subsamples efficiently, sparing valuable resources, time, and study personnel. The Winkelstein et al. (1987) classic study of young gay and bisexual men and the follow-up study by Osmond et al. (1994) both used multistage area probability designs. The sample for both studies was confined to a geographic area of the city that was defined by census tracts with the largest numbers of AIDS cases. The research interest was to determine the prevalence of HIV risk and HIV-related risk behavior of men residing in selected census tracts in San Francisco. Meyer et al. (2002) applied similar techniques in a random digit dialing (RDD) study to determine the feasibility of assessing sexual orientation among women residing in a selected ZIP code in an urban neighborhood in which there was an expectation of a high concentration of lesbians. Valleroy and her colleagues' (2000) study of HIV seroprevalence in young MSM is also technically an example of this type of study because once the researchers selected settings where young MSM were likely to be found, they conducted a time/location probability sample of all young men in those settings, independent of their sexual histories or identities.

6.3 Population of LGB Within Selected Settings or ZIP Codes

One limit to the population-based surveys described above is the relatively small size of the lesbian, gay, or bisexual sample that is generated, even sometimes when selected ZIP codes or settings are emphasized. Often these small numbers preclude more complex analyses of the interrelationship of relational, individual, and/or social behavioral variables. Only in general population surveys with large samples (e.g., 50,000) would there be adequate numbers for complex analysis. Conducting interviews with such a large number of respondents would be expensive. To overcome the small samples of LGB individuals generated in general population studies, researchers have designed prob-

ability studies that ensure the sample includes only respondents from the intended population. The Kanouse et al. (1991) study of gay and bisexual men in Los Angeles County was one of the early studies that screened potential respondents in an attempt to confine the sampled population to gay and bisexual men and to draw a large enough sample so more complex analyses could be achieved. Catania et al. (2001) extended the design to include three additional cities: San Francisco, New York, and Chicago. In both studies, the sample was drawn from ZIP codes that were selected based on their projected higher concentrations of gay and bisexual men. Data were collected by telephone using an RDD sample design. Because telephone number area codes and telephone prefixes are unique within ZIP codes in some cities, using ZIP codes to define the geographic boundaries in cities is a feasible, efficient sample design. However, because the overlay of telephone prefixes and ZIP codes are not always exact, researchers also query potential respondents regarding their residential ZIP code to ensure that the telephone number dialed was within the selected ZIP code. References (Kanouse et al., 1991; Binson et al., 1996; Blair, 1999) cited for both studies provide extensive details of the methods and sample designs used in these studies. The Diaz et al. (2001) and Mashburn et al. (2004) projects used a time/location probability design drawing, screening their samples of ethnic minority MSM from multiple sites where these men were known to congregate.

7 Suitable Use of Nonprobability Sample

Although only a probability sample allows us to make generalizations from the sample to the population, it was an experiment with a nonprobability sample that first challenged the medical model of homosexuality as psychopathology. In her revolutionary study, Evelyn Hooker (1957) acknowledged that her work was necessary because of years of bad sampling practice among researchers who sampled psychiatric and correctional institutions for their studies, resulting in skewed pictures of homosexuals as necessarily abnormal individuals with significant personality disorders. Hooker stated clearly why a probability sample was not necessary to address her research question: “for the present investigation the question is whether homosexuality is necessarily a symptom of pathology. All we need is a single case in which the answer is negative” (p. 30). By having expert judges evaluate responses to projective tests (Rorschach, TAT, MAPS) for 30 pairs of homosexual and heterosexual men (matched for age, IQ, and years of education), she could answer her question. Most of the men were viewed as functioning normally, and the judges could not do better than chance in identifying the homosexual men from their heterosexual counterparts. This is an example of an experimental design where the intention was to compare two samples that differ on (presumably) only one characteristic, not to project to some other population.

The health care literature is replete with numerous other examples of important studies conducted with nonprobability samples.

The reasons for this have to do with the scientific questions under investigation and the relative challenges of identifying the population to be sampled coupled with the high cost of probability designs. There can be good reasons not to recruit a probability sample but, rather, to draw a specific sample appropriate to the research question. For example, when studying a disease or a condition, researchers often have access to a limited number of patients who are all eligible for the study but who are not necessarily representative of the population. In such uses of nonprobability designs, the researcher is typically doing two things. First, the researcher is defining a population of interest for a particular project. That research population is usually representative of a larger population only in a limited way necessary for the research, perhaps on just one dimension. That is part of the reason one cannot generalize back to that larger population. The second thing the researcher is doing is using some low-cost method to obtain some members of the research population. With inappropriate use of nonprobability designs, step one is omitted or not recognized, leading to unsupportable claims of generalization. In addition to an acknowledgment that a given study has sample design limitations, a more difficult but useful extension of that recognition would involve a discussion of the ensuing biases and how the study results are affected.

Although there are various nonprobability sampling designs, there is no standard language for describing and categorizing nonprobability sampling, as terms become widely used in one field of research and not in others. For our purposes, we talk broadly about nonprobability samples as “convenience samples,” which is to say that the researcher samples individuals from the population who are easily available. The least systematic would be taking whoever comes along in whatever order they happen along until the desired number of individuals are selected. Each variation adds some level of systematic exclusion and inclusion criteria, such as seeking participants of the population of interest who provide a rich source of information that addresses the research question (purposive sampling) or selecting a set number of participants according to demographic and/or other categories to increase the variability in the sample, as Kinsey did (quota sampling), or having participants refer others they know to the study (snowball sampling). Obviously, these sample types are fluid, and researchers sometimes use combinations of approaches to achieve a sample that best suits their interests in achieving the research goals. Although in general more systematic recruitment appears to result in more diversity in the sample, it is nevertheless important to be aware of the significant limitations to generalizability from any convenience sample to the population of interest.

Determining the best approach requires thorough consideration of the research question and how the biases associated with a given convenience sampling strategy affect the interpretation of results. An often-cited example is the case of recruiting from gay bars for studies of alcohol use. If the research question involves understanding the subgroup of gay men at high risk for substance abuse or alcoholism, seeking out subjects from gay bars can be a purposive approach that might yield a large number of individuals at higher risk than the

general population of gay men. However, if the goal of the research is to determine the prevalence of substance use in the population of gay men, the bias present in a sample taken from bars would render the results useless.

In an example of purposive convenience sampling, Dilley et al. (2002) were interested in determining the efficacy of an experimental cognitive counseling intervention for high-risk men testing for HIV. Their sample, obviously, was purposive in that they had to select only high-risk men (defined as men who reported a previous testing history and high-risk sex with another man who was HIV-positive or of unknown status in the past 6 months). Although they asked every high-risk man in their program to participate—and in that sense they technically used a census sample of the program—it was still treated as a nonprobability sample because they could not generalize to the population beyond the testing program. Men who agreed to participate were randomly assigned to one of several groups, half of whom underwent the cognitive counseling intervention. Those who had the intervention were more likely than men who did not have it to reduce their risk behavior during the 12 months following testing. Although the study successfully measured the effect of the intervention, it would be inappropriate to take the descriptive data about levels of risk and characteristics of the highest risk men in the study and generalize from the study sample to high-risk men or to gay and bisexual men in general. The sample design was not constructed to answer such research questions. On the other hand, given that the sample was in effect a census, the characteristics and reported risk behavior can be generalized to high-risk men who seek HIV testing in this particular program. Even with this design, however, it would be prudent to determine if there were significant differences between those who participated and those who did not. In addition, further discussion of the ways in which the participants differ from other high-risk men, as best as can be determined, would help deepen the readers' assessment of the intervention.

7.1 Nonprobability Sampling for Pilot or Exploratory Studies

Nonprobability samples also can be appropriately utilized in pilot or exploratory studies. A conventional pilot study to test an instrument or interview protocol on a small number of participants usually provides enough variation in interviewing experiences and participant comments needed to improve the instrument. The only critical factor is that the participants in the pilot study have characteristics similar to those who will be interviewed in the main study; otherwise problematic questions may be missed. The same can be said for more formal pilots studies (e.g., those in which the interaction between interviewer and respondent is monitored and/or coded or those that explore respondent cognitive processes to determine participant comprehension and strategies used when responding to survey questions) (Presser & Blair, 1994). In the UMHS (Catania et al., 2001), several pilots were conducted to determine potential problems in item wording. In one of these pilot studies, a convenience sample of respondents was asked

questions about what the researchers thought might be potentially problematic question items. Each respondent was asked a survey question and then asked what the question meant to him, what difficulties he might have had in responding, and other comments he had about his perception of the intent of the question.

Sometimes the urgency of a public health crisis creates a need for rapid data collection that outweighs the limitations inherent in nonprobability samples. During the early period of the AIDS epidemic, as more and more gay men began coming down with various rare infections and immune problems it became critical to determine how gay men were responding to this epidemic. There was little time, as well as few financial or scientific resources, to consider drawing a random sample. The Multi-Center AIDS Cohort (MAC) study began recruiting as many gay men as were willing to complete a comprehensive set of questions every 6 months. This exploratory study was one of the first studies that tried to understand the relationship between psychosocial factors and gay men's attempt to change their sexual behavior in response to the epidemic (Emmons et al., 1986).

7.2 Nonprobability Sampling in Qualitative Research

Another broad area of research that uses nonprobability sampling to great advantage is qualitative research. Although qualitative studies cannot be generalized to the larger population, they allow researchers to explore the depth and richness of the LGB experience in ways that surveys often cannot. Frequently, the variables explored in surveys are first identified and developed through qualitative studies. Nevertheless, the sampling strategies for qualitative studies are often systematic, such as using purposive or theoretical sampling (i.e., selecting particular cases of the targeted population from which one can learn a great deal about issues of central importance to the purpose of the research).

Gartrell and colleagues provide an excellent example of using various convenience-sampling techniques to acquire a sample for a qualitative study of lesbian mothers (Gartrell et al., 1996, 1999, 2000). From 1986 to 1992, they recruited study participants who were either pregnant or in the process of inseminating, using "informal networking and word of mouth referrals . . . announcements at lesbian events, in women's bookstores and in lesbian newspapers" (p. 274) in the three cities where the investigators were located (i.e., Boston, Washington, DC, and San Francisco). Although they did not report a preset quota for numbers of women of color to include, they distributed "flyers at events for women of color" to ensure diversity of racial/ethnic background within the sample (p. 274). Using these techniques, the study team recruited 84 families (39 from San Francisco, 37 from Boston, 8 from Washington); 14 of the families were headed by single mothers, and the remaining families were headed by two mothers. This is an excellent example of applying purposive, snowball, and quota sampling procedures to ensure a large and diverse sample from which to identify and understand important issues of lesbian motherhood.

The above examples are not meant to be exhaustive, as the possibilities for useful studies with nonprobability samples are virtually endless. It is also worth noting, however, that for some populations nonprobability sampling may be inadequate and probability sampling actually more efficient for identifying otherwise difficult to reach populations or recruiting individuals of interest. For instance, we have proposed studying men at exceptionally high-risk for HIV; that is, from a public health perspective, the “core” members of sexual networks who are at significant risk both of acquiring and spreading HIV and other sexually transmitted diseases. The reason that probability sampling can be useful for identifying these core members is that we know from other studies that core members do not respond to standard nonprobability recruitment efforts. Martin et al. (2003) looked at sampling effects comparing a probability sample to a snowball sample. They found that high-risk men recruited into a probability sample were less likely than low-risk men to refer other men into the snowball sample, and that high-risk men who did refer men into the snowball sample were less likely to refer other high-risk men. Therefore, it is likely that we would have difficulty recruiting these “core” men through convenience sampling. Fortunately, core men can be and have been recruited reliably through probability sampling. Thus, probability sampling appears to be a good method for recruiting the target population.

8 Emerging Sampling Designs

8.1 Respondent-Driven Sampling

Thompson and Collins (2002) identified a category of sampling designs they termed “link-tracing” designs, which they defined as “Any design in which links or connections between units are used in obtaining the sample.” Of recent interest is a link-tracing design called respondent-driven sampling (Heckathorn, 1997), which if its assumptions hold can produce a probability sample.

In respondent-driven sampling, a sample is collected using a chain-referral procedure. That is, respondents are not selected from a sampling frame, but are selected from the social network of existing members of the sample . . . the sample is used to make estimates about the social network connecting the population. Then, this information about the social network is used to derive the proportion of the population in different groups (for example, HIV+ or HIV–) [Salganik & Heckathorn, 2004].

This design has been proposed primarily as a method of sampling “hidden” populations.

For the estimation procedure to be unbiased, several assumptions must be met. It is assumed that the “seeds,” who are the first wave of respondents, are drawn with probabilities proportional to their network size (i.e., the number of other people in the network to whom

they are linked), although this condition is sometimes relaxed. It also is assumed that respondents recruit randomly from the other people in their network to whom they are linked. Finally, it is assumed that respondents know the number of people in the target population to whom they are linked (i.e., the number of people they might possibly have referred into the study, or their “degree”).

Whether these conditions hold in a particular application is an empirical question. Certainly, it is not difficult to suggest factors that might undermine them. Because there are no rules for selecting the seeds, it is possible that their chance of selection is not proportional to the size of their network or degree. For example, if seeds are selected because they are somehow known previously to the researcher or through an advertising mechanism such as flyers, their chance of selection is no longer proportional to the size of their degree. Even if the chance of their selection is *related* to degree (i.e., that people with larger degree in general are more likely to be selected than those with smaller degree), this is still not the same as “proportional to degree.” That is, a person with degree 10 may be more likely than someone with degree 2 to be selected as a seed, but perhaps not five times as likely.

Recruitment involves, in the main example given in the Salganik and Heckathorn article (2004), passing a coupon to those who are recruited and receiving payment for recruitment of participants. It does seem likely that respondents might favor recruiting those network members they think are more willing to participate in the study, rather than selecting them at random. That is, one rational recruitment strategy would be to achieve the highest (or quickest) payoff for the least effort. When this occurs, it violates the assumption that participants will recruit randomly from the other people in their networks.

One final concern is whether participants would really know the number of people in the target population to whom they are linked. Of course, some may know, whereas others may not. A reasonable scenario would be that those with small personal networks could report the number more accurately than those with large networks. The counting task is simply easier for those with tiny personal networks. If this were the case, a systematic reporting bias would result.

This is a complex design (only partially described here) that depends on a number of assumptions. The key point is that for some studies the assumptions may hold well, whereas in other situations they do not. Prior to deciding whether to use this sampling strategy, investigators should attempt to make some assessment of the extent to which the circumstances of their study would result in significant departures from these assumptions. This possibility too, however, would vary in difficulty from case to case.

In contrast to more basic designs, respondent-driven sampling, such as network sampling, relies to a greater degree on certain conditions and respondent behaviors. As such, they carry greater risks of the type noted for complex sampling designs—most seriously the loss of a statistical basis on which to make sample estimates of population para-

meters. However, if future work shows that the assumptions of respondent-driven sampling are not as easily violated as they might seem, or that the estimates produced are robust to violations of those assumptions, respondent-driven sampling may provide a more economical way of drawing probability samples with difficult-to-reach populations. The use of pilot studies, if feasible, may be the wisest course.

8.2 Web-Based Surveys

The web (Internet) is an attractive data collection mode for a variety of reasons, cost being a primary one. It permits a level of administrative control, such as skip patterns and edit checks, that are not possible in noncomputerized modes, such as conventional mail. There are many possibilities for incorporating multimedia technologies in web surveys, although they are not yet widely used in survey research applications. It is important, however, not to allow the utility of web technologies as a *survey administration mode* to eclipse the limitations inherent in its use as a *sampling strategy*.

Although the web is essentially an alternative means of data collection and not a sampling procedure, its effective use (outside of list samples or nonprobability samples) often requires dealing with multimode designs, which do have special sampling issues. Although multimode designs in general are beyond the scope of this chapter, we believe it useful to address some of the specific sampling issues that may, under some circumstances, permit effective use of the web. The major drawbacks to web surveys involve sampling and response rates. These obstacles are often sufficiently severe to render proposed research impractical. However, there are some circumstances under which the web, alone or in conjunction with another mode, may produce acceptable results.

One might consider several broad categories of web surveys. First, a population may be defined by virtue of web access (e.g., members of a listserv or some other type of online group). Second, the web may simply be a convenient mode for reaching a narrowly defined, special population, such as members of a professional association or subscribers to a magazine for whom e-mail contact lists are available. Third, the target population may be simply a subgroup of the general population.

The first two types of survey are relatively straightforward. Assuming that the e-mail list is available, they present no special problems beyond those that apply to sampling from any other list. The use of the web for conducting general population surveys, and by extension subgroups within the general population, is far more problematic.

First, only about 40% of the U.S. population is currently on the Internet (Newburger, 2001). Even if a list of this population were available, which it is not, the degree of undercoverage would make the web inappropriate for serious research (though it might be acceptable for some types of instrument testing, developing hypotheses, and other

applications where either nonprobability samples or seriously flawed probability samples are useful).

At present, utilizing the web for general population surveys seems feasible only as part of a dual-frame design. It is worth considering the savings that might be available from dual-frame designs in which web-based data collection is used for web-accessible members of the target group, and telephone data collection is used for those who are not in the web frame. Web-phone designs might be particularly cost-effective for reaching rare groups; and if so, we need to understand those circumstances.

Screening costs on the web should be lower than on the phone because no incremental labor is needed to contact potential respondents. This reduction in cost is not dependent on geographic clustering in the target group; therefore, web-based data collection may offer savings where TCS cannot. Also, the inherently lower cost of web-based screening should be increasingly important as the group becomes more rare, so web-based sampling may be useful for very rare groups even if they are geographically clustered and hence amenable to TCS and/or stratification.

For telephone interviewing, if the target group has zero tendency to cluster geographically, RDD with no clustering is the best option. If the target group has a nonzero tendency to cluster geographically, TCS is preferable (e.g., with list-assisted RDD at the first stage and a cluster size of 3).

The question now becomes how can web sampling be combined with one of these telephone alternatives to improve the cost efficiency of the sampling plan?

Web-based data collection can be accomplished in three main ways. First the questionnaire may be posted on some web site for any and all who might want to participate. Obviously, this method generates a pure sample of volunteers and is not acceptable if our goal is to achieve a probability sample. There is no way to know the inclusion probabilities of a sample obtained in this manner.

With the second method a sample is drawn from a panel of web users who have agreed to participate in research projects: Examples include the panels maintained by Harris Interactive, Ipsos Research, and more recently Survey Sampling. Because the members have explicitly agreed to participate in research, usually with some system of incentives, panel vendors are often able to estimate the level of response—30% or so is not uncommon.

The third method utilizes a sample drawn from a list of “opt-in” users. These sample members are people who, in the process of using the Internet, have agreed to receive e-mail communications. Such lists are constructed from multiple sources, and their quality is far less clear than a panel. Because opt-in list members have agreed to receive communications, solicitation to participate in a survey does not violate anti-“spam” restrictions. However, they did not explicitly agree to respond to survey solicitations, and their response rate is correspondingly low, often in the single digits. However, it is important to note that this frame is generally much larger and provides greater population cov-

erage than does a panel. Both of these characteristics are important issues when sampling rare populations.

Essentially, implementing this design becomes a problem of optimal allocation between a telephone stratum (where coverage is high, but the costs of screening are as well) and a web stratum (which has poorer coverage but very low costs for screening). The detailed calculations to determine optimum allocation are beyond the intent of this chapter (see Blair & Blair, 2005, for a full treatment). However, even though the correct use of the dual frame approach would require consulting with a sampling statistician, it is useful to understand the general issues involved.

Blair and Blair (2006) conclude that cost savings may be available from dual-frame web-phone designs (versus all phone), but the situation is complex and is not entirely free of some of the basic drawbacks noted at the outset of this discussion. The main finding was that web-based data collection with web panels might offer substantial savings for rare groups even when there is some clustering in the population.

Given the cost benefits that might be available from web-based data collection for rare groups, should it be used? The answer, of course, depends on one's willingness to treat a sample of opt-in users or a sample of panel members, knowing the potential for coverage bias, as representative of the larger online population. This decision sometimes must be made in the context of being able to do the survey or not, given a fixed budget.

A theoretical solution to this problem is to stratify the general population on the basis of "in frame vs. not in frame" rather than "online vs. not online." However, there is the practical difficulty of getting telephone respondents to give an accurate indication of whether they appear in any given online frame so they can be sorted out. Overall, there are two practical responses to the issue of coverage bias in the online frame. One is to use the biggest possible frame, actively draw the sample, and take a leap of faith that the result is representative (and do likewise regarding the issue of nonresponse bias). Second, one may try to estimate the bias by comparing the telephone and online results. If the bias can be estimated, one has the possibility of correcting for it while retaining the cost savings that led the researcher down this design path originally.

For some research goals, the possible cost advantages of web-based data collection in reaching rare groups may be sufficient to overcome its theoretical drawbacks; in other cases it may not. What is important is that the decision be informed and that the potential implications for the accuracy of the survey results be understood.

In conclusion, even if web-based data collection is cheaper than telephone interviewing, it is not desirable to rely exclusively on web-based data unless the target group is limited to Internet users because of the potential for coverage bias in the sample. The more defensible approach is a dual-frame design in which web-based data collection is used to gather data from web-accessible members of the target group, and telephone interviewing is used to reach the others.

9 Summary

Having faced the AIDS epidemic more than 20 years ago with few reliable data about the prevalence and sexual practices of gay men in the United States, it is exciting to see so much growth in the health research area where probability sample designs are used to describe the LGB populations. Clearly these studies (and others like them not presented here) provide far more reliable information about these rare and elusive groups than had been known in all the years of research on homosexuality that preceded them.

The detailed discussions included in this chapter provide only a basic introduction to probability sampling. The information presented is intended only to provide a description of the principles underlying probability sampling, why they are important when designing reliable surveys, and some sense of the range of available designs. Additional knowledge is required to develop appropriate sample designs for specific applications. Quite a bit more training and experience is necessary, and someone with sufficient expertise and experience should be sought out to provide consultation or to collaborate in the development and implementation of a study's sample design. It is important to keep in mind how much sampling interfaces with the data collection operation in many of the more complex sample designs. If a research question is worth asking, it is certainly worth generating an appropriate sample so the time, effort, and resources involved in collecting and analyzing the research data can provide reliable answers to the research questions proposed. Certainly more studies using well thought-out sample designs will do much to improve our knowledge of, and efforts to improve, the health and well-being of LGB populations.

Finally, the understandable pressures to conduct studies that produce useful substantive results have made it difficult to also conduct the survey methodological research needed to improve the design of LGB surveys. Until more methodological research is done, large uncertainties will remain about how well different designs will perform in particular circumstances. These uncertainties underlie this chapter's emphasis on expert consultation, careful planning, and the use of pilot studies as the bases for successful sample design.

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Further Reading Related to Sampling Methods

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- Czaja, R., and Blair, J. (1996, 2005) *Designing surveys: a guide to decisions and procedures*. Pine Forge Press, Thousand Oaks, CA. *Book includes an extensive discussion on survey sampling for readers’ reference (chapters 7 and 8).*
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a special issue of *Public Opinion Quarterly* celebrating 50 years of survey research. The article provides an excellent overview of the history of survey sampling in the United States.

Kalton, G. (1983) *Introduction to survey sampling*. Sage, Newbury Park, CA. One of the Little Green Sage series and an excellent resource for someone interested in learning about probability sampling.

Kalton, G. (1993) Sampling considerations in research on HIV risk and illness. In: Ostrow, D.G., and Kessler, R.C. (eds) *Methodological issues in AIDS behavioral research*. Plenum, New York. Chapter covers many basic sampling issues for surveys related to HIV risk and illness.

Sudman, S. (1983) Applied sampling. In: Rossi, P.H., Wright, J.D., and Anderson, A.B., (eds) *Handbook of survey research*. Academic, San Diego, pp. 145–194. An overview and practical guide to survey sampling taking into consideration limited resources. It includes many interesting and useful real world examples. Material for this chapter was obtained from the book *Applied Sampling Academic*, San Diego, 1976.

Sudman, S., Sirken, M.G., and Cowan, C.D. (1988) Sampling rare and elusive populations. *Science* 240:991–996. An excellent overview to sampling hard to reach populations.

National Professional Associations of Survey Research and Sampling

American Association for Public Opinion Research. Website: <http://www.aapor.org/>.

American Statistical Association, Section on Survey Research Methods. Website: <http://www.amstat.org/sections/SRMS/index.html>.

National Survey Research Centers

Abt Associates, Cambridge, MA

Mathematica, Princeton, NJ

National Opinion Research Center (NORC), University of Chicago, Chicago, IL

Rand, Survey Research Group, Santa Monica, CA

Research Triangle Institute (RTI), Research Triangle Park, NC

Survey Research Center, Institute for Social Research, University of Michigan, Ann Arbor, MI

Westat, Rockville, MD